

Guru Sakthi S

Electronics & Communication Engineering Undergraduate

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering undergraduate with a strong foundation in embedded systems, digital electronics, and IoT solutions. Experienced in C programming, hardware–software integration, and engineering project development with hands-on implementation skills.

EDUCATION

B.E. Electronics and Communication Engineering

Kongu Engineering College

CGPA: 8.92

2023 – 2027

Higher Secondary Education

Bharathi Vidya Bhavan

Percentage: 85.33%

2022 – 2023

SKILLS

Technical Skills: Microcontrollers – 8086, 8051, ESP32, PIC16F877A, Raspberry Pi-4; **OS** – Linux (Ubuntu 24.04 LTS);

Version Control – Git, GitHub; **Languages** – C, Java, C++, Python; **Tools** – Proteus, MATLAB, VS Code.

Soft Skills: Problem Solving, Team Collaboration, Communication.

AREAS OF INTEREST

Embedded Systems | C and Java Programming

PROJECTS

Indian Sign Language (ISL) Translator – Bidirectional

(Aug 2025 – Sep 2025)

- Developed a bidirectional ISL translator enabling real-time communication between hearing-impaired and non-hearing-impaired users.
- Achieved **92% classification accuracy** using deep learning models trained on custom gesture datasets.

Smart IoT-Based Waste Segregation and Monitoring System

(Nov 2025)

- Developed an IoT-based automated waste segregation system using multiple sensors for real-time identification.
- Implemented cloud-based data storage and analytics using Firebase, enabling live monitoring, visualization, and area-wise waste tracking through map-based location mapping.

Smart Air Purification System for Small-Scale Industries

(Apr 2025)

- Built an automated air quality monitoring and purification system using MQ gas sensors for metropolitan areas.
- Implemented dual-stage activated carbon filtration with microcontroller-based control.

AI-Based Automated Ticket Management System

(Mar 2026)

- Designed and developed an AI-driven ticket management system using **Natural Language Processing (NLP)** to analyze ticket content, urgency, priority, and emotional sentiment.
- Implemented **BERT-based transformer models** for automated ticket classification and priority prediction.

ACHIEVEMENTS

- First Place** – C Programming Competition for problem-solving and algorithmic efficiency.
- Top 5 Winner** – EBox Top Coders (C) national-level competitive programming contest.
- Academic Excellence Scholarship** – Awarded **20,000** for outstanding academic performance.
- YUKTI Innovation Challenge 2025** – Presented an innovative engineering project at national level.
- Second Prize** – Simulation for embedded system simulation using **Keil Software**.
- Third Prize** – 30-hour Software **AI Hackathon** for AI-based solution development.

CERTIFICATIONS

- Oracle Java Foundation Associate (1Z0-811)** – Certified in core Java programming fundamentals.
- Advanced Embedded System Design** – STM32 and RTOS based industrial training certification.
- Industry Ready Engineer** – Sunshiv program focused on industry-oriented engineering skills.

LANGUAGES

Tamil (Native) | English (Professional)